

Nicholas J. Fettig

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EDUCATION

Swarthmore College | Swarthmore, PA

B.S. in Engineering and B.A. in Computer Science | GPA: 3.82/4.0

May 2026

Relevant Coursework: Artificial Intelligence, Computer Systems, Compilers, Computer Vision, Data Structures & Algorithms, Mobile Robotics, PDEs, Probability, Stochastic & Numerical Methods

Awards: Martin S. Kapp Scholarship awarded by the Society of American Military Engineers

TECHNICAL SKILLS

Languages: Advanced in NodeJS, Typescript, Python, MatLab, HTML/CSS, C, C++; Proficient in OCaml, SQL, Java, Golang

Technologies: Lit.js, React.js, Next.js, Docker, Kubernetes, AWS, Git/Github, Linux Administration, ROS2, LaTeX

WORK EXPERIENCE

Reliable Robotics | Mountain View, CA

May - August 2025

Software Engineering Intern

- Designed, developed, and tested mission-critical display systems for remote operation of automated aircraft
- Lead development of "Big Board," a large display in the Mission Control Center used by remote airplane pilots
- Integrated tracking for 8 Reliable Robotics aircraft using ADSB data, developed in Lit.js and deployed with AWS EC2

Carnegie Mellon University | Pittsburgh, PA (Hybrid)

May 2024 - May 2025

Computer Science Undergraduate Researcher

- Researched deployment challenges for Reinforcement Learning-based Traffic Signal Control (TSC)
- Organized 200+ works in the RL-based TSC field and sorted into a novel challenge classification taxonomy

Swarthmore College | Swarthmore, PA

August 2023 - Present

Teaching Assistant

- Run weekly study sessions for Intermediate Python and C courses for CS & Engineering Departments
- 100% satisfaction rate from advisor for CS courses and Lead TA for Intro. to Computer Engineering

Graphicient | North Bethesda, MD

May - August 2023

Software Engineering Intern

- Refactored a dashboard used to track CCRI network-security compliance for Graphicient's largest client, Three Wire Systems, introducing notion of a "Cyber Score" and increasing cybersecurity-assessment accuracy by 25%
- Developed NextJS web-app running ExpressJS microservices with secure endpoints, all scaled using Kubernetes

PROJECTS & COMMUNITY LEADERSHIP

Home Server | Docker, Linux Administration, Cloudflare

December 2025 - Present

- I run an Ubuntu server at home to host containerized services like my personal site and Jellyfin
- Securely exposed ports on my machine using key-only SSH access and Cloudflare tunnels

Swarthmore College Computer Society | Next.js, Docker, Linux Administration

January 2022 - Present

Vice President

- Lead development of student directory and mailing list web-app services developed in Next.js
- Maintain club-owned servers running Proxmox, various Linux VMs, dockerized services

Montgomery County Covid-19 Vaccination Website (MoCoVax) | HTML/CSS, Node.js

April 2021

- Developed and deployed a website with 120,000+ visits to show available vaccine appointments